

Gesture Controller using Hand Tracking

README / Setup and Usage Guide

A Python project that uses a webcam, MediaPipe hand landmarks, OpenCV, and PyAutoGUI to control the computer mouse with hand gestures.

Features

- Real-time hand tracking
- Index-finger cursor control
- Thumb-index pinch used as a click
- Cursor smoothing to reduce jitter
- Live webcam feedback
- Press Q to exit

Requirements

Python 3.x
Webcam

```
pip install opencv-python mediapipe pyautogui
```

How to Run

1. Save the supplied code as `gesture_controller.py`.
2. Open a terminal in the project folder.
3. Install the packages shown above.
4. Run: `python gesture_controller.py`
5. Move the index finger to control the cursor.
6. Bring the thumb and index fingertip together to click.
7. Press q while the camera window is active to stop.

How It Works

MediaPipe returns normalized hand landmarks. Landmark 8 is the index fingertip and landmark 4 is the thumb tip. The index coordinates are mapped to the monitor resolution. The Euclidean distance between the two fingertips is then checked; if it is below 30 pixels, `pyautogui.click()` is called.

Libraries

Library	Purpose
OpenCV	Webcam capture, frame processing and display
MediaPipe	Hand detection and landmark tracking
PyAutoGUI	Operating-system mouse movement and clicking
math	Fingertip distance calculation

Important Notes

- The program controls the real OS mouse.
- PyAutoGUI FAILSAFE is disabled in the supplied code.
- The current pinch condition may cause repeated clicks if fingers remain close.

- Lighting and camera position affect tracking accuracy.